Español | **English**

Gas detectors for H_2S, CO and fuel vapors

Options for an industrial site where each detector works **on its own**: no central controller, with a local light (strobe), a horn and, where possible, a screen showing the live gas level. The list covers all-in-one units and one-detector-per-gas setups, in mid and high price ranges.

H_2S (hydrogen sulfide) | CO (carbon monoxide) | Fuel vapors = combustible gas, % LEL | Standalone: strobe + horn + display

Technical terms? See the glossary ↓

Prices are not quotes. They are what US/UK resellers listed online in October 2026, before tax, shipping and installation. Most of these units are built to order, and the exact sensor setup changes the price. Ask an authorized distributor for a quote with your gases and ranges before buying.

Our case

The site

- **Administrative office, unclassified area**: general-purpose equipment is fine.
- About **1 km away there is a port** where fuel and crude oil tankers unload; odors reach the office.
- Goal: **local alarm** (visual and audible) when levels get high, with no central panel.

Chosen sensors

- **H_2S**, in the lowest range available, to get closer to odor levels.
- **CO**, at breathing height.
- **Infrared LEL**, as a safety backstop. At 1 km it will normally read 0; it would only rise with a strong vapor cloud, and that's not a fault.
- Optional: **PID** for fuel and crude vapors in ppm/ppb, the most sensitive sensor for these odors.

Quotes (Oct 19–27, 2026)

- Manufacturers: Sensidyne, Honeywell, Crowcon.

- US distributors: PK Safety, FLW, Instrumart.
- **Sensidyne quoted** (Oct 21, received Oct 27) **2 SensAlarm FLEX units**: unit 1 = CO 0–500 ppm + remote **H_2S 0–10 ppm** mounted low; unit 2 = IR LEL. Equipment **$11,354** + calibration kit with gases **$1,770 = $13,124**. Built-in 95 dB horn, strobes and display; plugs straight into 110/220 VAC. 3-year sensors, 2-year warranty, 2–4 week lead time, Net 60 export terms (subject to approval), valid to ~Nov 20. Optional 0–5 ppm PID (the only low-range PID quoted): **$8,572**; it can false-alarm because it isn't gas-specific. Pending: a lower CO range, boxed weight, and whether the kit can ship without cylinders (dangerous goods by air).
- **Crowcon quoted** (Oct 22–23) the Xgard Bright: one unit per gas, no panel, with relays. Chosen setup, all A3 versions (M20 thread with a ½" NPT adapter included): **CO 0–50 ppm $1,331 · H_2S 0–25 ppm $1,331 · IR LEL propane $1,780 · calibration kit $208.95 = $4,650.95**. The built-in sounder is only 40 dB, so an external horn and strobe is needed: Crowcon's integrated accessory (price pending) or a generic one (~$100–180), plus a 24 VDC supply (~$50–70). **No low-range PID**, so it's dropped. The kit works for the 25 ppm H_2S; Crowcon doesn't sell calibration gas, so it's best bought in Venezuela. Spare sensors (CO 0–50 ppm S011572/S, H_2S 0–25 ppm S011257/S): $700.40 each, ~2-year life (buy near year 2). Shipping: 8 lb and 10 × 6 × 5 in per detector, boxed (all 3 ≈ 24 lb / ~11 kg), with our own carrier. 8-week lead time, prepayment, quote valid to ~Nov 21. Pending: part number, price, dB and

power draw of the horn/strobe accessory (asked again Oct 23). Quote report for approval (PDF, Spanish) ↗

- **Comparison:** with calibration gas in both, Crowcon ≈ $6,000–6,100 vs. Sensidyne $13,124 (≈ 2.2×). Recommendation: **Crowcon** for the 3 gases; to catch the odors, add only Sensidyne's PID (≈ $14,600–14,700). Freight with Sandoval (Miami → San Cristóbal, $7.50/lb): ≈ $200 Crowcon, ≈ $250–330 Sensidyne (estimate, final rate pending). Crowcon vs. Sensidyne comparison (PDF, Spanish) ↗
- Others: pending.

Short answer

BEST ALL-IN-ONE, PERMANENT

Sensidyne SensAlarm FLEX

Two wall units (up to 2 sensors each), with a color screen, built-in strobes and a horn. Configuration proposed by Sensidyne for our case.

BEST PER-GAS, MID-HIGH

3 × Crowcon Xgard Bright

Use the version with the built-in light and sounder, and an infrared sensor for fuel vapors.

LOWEST COST (NON-HAZARDOUS AREA)

Macurco 6/12 series

One unit per gas, each with a display and buzzer, plus a strobe wired to its relay.

NO WIRING / TEMPORARY

Industrial Scientific Radius BZ1

A battery area monitor with a screen, a 108 dB horn and strobes. You can move it around the site.

First: is the area classified?

Where fuel vapors can build up, the area is often classified as hazardous (Class I Div 1/2 in the US, ATEX/IECEx zones elsewhere). In a classified area, the detector, horn and strobe all need the matching explosion-proof rating. The cheaper "general purpose" units below

(Macurco, Forensics) are **not** for classified areas. Your safety officer or the plant's area classification drawing settles this, and the answer decides which half of this page applies.

Our case: unclassified (administrative office). No explosion-proof equipment is needed, so the general-purpose options also apply.

All-in-one (several gases in one unit)

Sensidyne SensAlarm FLEX

HIGH

Fixed, wall-mounted, permanent

7" color LCD | Built-in LED strobes | 95 dB horn | 4–8 relays | 4–20 mA / Modbus

`Quoted: $13,124 for 2 units with calibration kit and gases (Oct 2026)`

GOOD

+ Closest match to the brief: each unit shows its gases on a big screen and alarms on its own, with no central needed.
+ With 2 units there is a spare channel to add a PID (fuel and crude vapors).
+ The sensor heads are rated Class I Div 1, so they can go in the hazardous spot.
+ Spare relays can drive extra strobes, fans or valves later.

BAD

– **Up to 2 sensors per unit** (confirmed by Sensidyne): H_2S, CO and LEL need 2 units.
– The main display is rated only Class I Div 2, so it must sit in a less hazardous spot (not an issue in our case, unclassified area).

Industrial Scientific Radius BZ1

HIGH

Transportable area monitor

LCD | 108 dB horn | Red + blue strobes | Battery | Up to 7 gases

`≈ $7,400 – $14,100 (4-gas LEL / CO / H₂S / O₂)`

GOOD

+ Very loud and very visible; built for loud industrial sites.
+ No wiring: about 7 days typical on battery, about 30 days with the external power supply.
+ Can add a PID sensor for diesel or other heavy fuel vapors.
+ Hazardous-location rated.

BAD

– Expensive per unit, and the docking station is extra.
– Made for temporary use; someone has to charge it or keep it on external power.
– The "with pump" models drain the battery about twice as fast.

- The most expensive fixed option: $13,124 quoted, about 2.2× the Crowcon price.
- A 95 dB horn is quieter than the area monitors; add a horn on a relay in a noisy plant.

Sensidyne product page ↗

SensAlarm FLEX brochure (PDF, 8.7 MB) ↗

Application notes for our site (PDF) ↗

H_2S 0–10 ppm sensor (PDF) ↗

Infrared LEL sensor (PDF) ↗

Moisture barrier (PDF) ↗

SensFlex PID, optional (PDF, 0.9 MB) ↗

PID horn 7017380 (PDF) ↗

BuyGasMonitors ↗ Airgas ↗ Tenaquip ↗

Honeywell BW RigRat

HIGH

Transportable area monitor

LCD | Audible + visual alarm | Battery | LEL / CO / H_2S / O_2

≈ $9,400 – $11,050

GOOD

+ Same idea as the Radius: one rugged box, loud alarm, no wiring.
+ Honeywell service network and calibration gas are easy to find.

BAD

- Costs more than the Radius in most listings.
- Many configurations; check that the exact part number has the sensors you need.
- Built to order, so lead times can be long.

Blackline G7 EXO

HIGH

Area monitor + cellular alerts

LCD | Audible + visual alarm | 100-day battery | Cellular | H_2S / LEL-IR / CO / O_2

Subscription (device + service plan), quote only

GOOD

+ About 100 days on one charge, with the longest-life sensor setup.
+ Sends alarms to phones over cellular, so people hear about it even off site, still without a central panel.
+ Uses an infrared LEL sensor, which H_2S can't poison.
+ 3-year hardware warranty.

BAD

- Monthly or yearly service fee; the remote features stop if you stop paying.
- Needs cellular coverage at the site for the remote alerts.

Zoro ↗ Airgas ↗

User manual (PDF, 3.2 MB) ↗

AllSafe Industries ↗ Datasheet (PDF) ↗

Forensics Detectors 4-Gas Wall Mount (FD-60-4GAS)

BUDGET

Fixed, general purpose

CO / H_2S / O_2 / LEL · Relay output · Wall mount

≈ $1,395

GOOD

+ By far the cheapest way to get all three gases in one wall unit.
+ The relay can switch an external strobe or fan.

BAD

– Not for classified (explosive) areas.
– Smaller brand with thinner support and spare-sensor supply.
– Online listings don't confirm the display or alarm specs; check the datasheet before buying.

WC Safety ↗

One detector per gas (you need 3)

You install three heads: H_2S, CO and LEL. Each one shows its own reading and alarms on its own. This costs more up front than one box, but you can place each sensor at the right height for its gas (see the tips below), and one failed sensor leaves the other two working.

Crowcon Xgard Bright

MID-HIGH

Fixed, explosion-proof, per gas

MSA Ultima X5000

HIGH

Fixed, explosion-proof, 1–2 sensors per head



Xgard Bright H_2S (front and display)



Xgard Bright LEL



Side view: sensor at the bottom



Stainless steel version (reference)



D2x horn + strobe (Option A)

Photos: Crowcon and TriNova. The LEL photo shows the MPS sensor; the quoted infrared unit uses the same housing and display. CO detectors look the same: only the label and reading change.

Display | Built-in beacon + sounder option | Relays | 4–20 mA / Modbus | IR LEL available

≈ £740 – £985 per head (≈ $950 – $1,300), ×3 · Crowcon quote (Oct 22-23): $1,331 per CO or H_2S head · IR LEL propane $1,780

GOOD

+ Order it with the integrated audio-visual alarm and each head is a complete standalone station: screen, light and sound.
+ Flameproof, so it works in classified areas.
+ Wide H_2S ranges (10–200 ppm), and an infrared LEL sensor for fuel vapors.
+ Best price for that feature set among the big brands.

BAD

– Mostly sold through UK/EU distributors; fewer US stockists.

Display | Relays | Bluetooth app | XIR+ infrared LEL | 2 sensors per head

≈ £3,010 – £5,125 per head (≈ $3,900 – $6,600)

GOOD

+ Two sensors in one head, so two heads cover all three gases (e.g. H_2S + IR LEL, and CO).
+ Top-tier sensors with long life; the Bluetooth app makes setup and checks easy.
+ Strong US support network.

BAD

– No built-in horn or strobe: add an explosion-proof horn/strobe wired to its relay.
– Most expensive per head.

- The beacon/sounder add-on is often out of stock; confirm the lead time.
- The built-in sounder isn't as loud as a dedicated horn in a very noisy plant.

Frontline Safety ↗ Rockall Safety ↗

Beacon/sounder add-on ↗

Product & installation manual (PDF, 1.9 MB) ↗

Multilingual user manual (PDF, 1.6 MB) ↗

User manual (PDF, 1.6 MB) ↗

Data sheet (PDF, Spanish, 5.2 MB) ↗

UL dimension drawing (PDF, 0.2 MB) ↗

Frontline Safety (dual) ↗ Rockall Safety ↗

Honeywell Sensepoint XCD

MID-HIGH

Fixed, explosion-proof, per gas

Backlit LCD | 2 alarm + 1 fault relay | 4–20 mA / Modbus

`≈ $2,250 – $3,100 per head, ×3`

GOOD

+ Proven, widely stocked; you can set it up without opening the housing.
+ The relays can drive a horn/strobe directly, with no controller.
+ Replaceable sensor cartridges (H_2S ≈ $570).

BAD

– No built-in horn or strobe; budget for three explosion-proof horn/strobes.
– Honeywell added a 16% surcharge in 2025, and lead times can run 2+ weeks.

CTI GG series + CTI horn/strobe

MID

Fixed, per gas; explosion-proof (EXP) versions

Display | Relays | 4–20 mA | Matching horn/strobes

`GG-H2S ≈ $1,120 – $1,600 · horn/strobe ≈ $260 – $780`

GOOD

+ US maker that sells the detector and the matching horn/strobe together, so they're easy to pair.
+ The EXP version is made to drive a horn/strobe and ventilation directly.
+ Mid-range price with an online store.

BAD

– The horn/strobe is a separate box (more wiring).
– Less known outside North America.

Kele (H_2S) ↗ e-Controls ↗ Instrumart ↗

CTI online store ↗

GG-H2S-EXP at Keep Supply ↗

RKI Beacon 110 kit

MID-HIGH

Explosion-proof horn + strobe with a one-sensor local controller

Horn + strobe built in | Display | 12 VDC option

`Varies by kit; see Transcat`

GOOD

+ Explosion-proof horn and strobe built into one package.
+ Each kit runs one sensor locally, so it's still standalone with no plant-wide central.
+ Can run on 12 VDC, which helps where mains power is awkward.

BAD

– You still need three kits (one per gas) plus the sensors.
– Bulkier install than a single head.

Transcat ↗

Macurco 6 / 12 series

BUDGET

Fixed, general purpose, per gas

Digital display | Internal buzzer | Alarm relay → strobe | Fan relay

`H2S TX-6-HS ≈ $865 · CO CX-6 ≈ $415 · LEL GD-6 ≈ $550`

GOOD

+ Lowest cost for three real standalone detectors (about $1,850 for all three).
+ Each one has a display and a buzzer; the relay runs a strobe and an exhaust fan.
+ Simple to install and maintain.

BAD

– Not explosion-proof: only for non-classified areas (shops, garages, battery rooms).
– The buzzer is small; add a strobe/horn for a loud plant.
– Catalytic LEL sensor; see the H_2S warning below.

PK Safety (H_2S) ↗ 123 Security (LEL) ↗

123 Security (CO) ↗ BuyGasMonitors ↗

Other options reviewed: Honeywell Midas S2 and Sensepoint XRL

Checked against the requirement: standalone detectors, no central panel, visual and audible alarm and, if possible, an LCD screen. Neither is the best choice for wall-mounted detectors; here's why.

Honeywell Midas S2

SPECIAL CASES

Sampling detector: draws air through tubing from up to 30 m away, one per gas

2.4" color display | Alarm LEDs | Relays | 4–20 mA / Modbus TCP | 24 VDC or PoE

`≈ $1,780 per unit + $448–680 cartridge, + 16% surcharge; > $7,000 for all 3 gases`

PROS

+ Color display and bright LEDs; relays can drive a horn/strobe, so it works without a central panel.
+ Good for watching pits, tanks or hard-to-reach spots without putting the unit in the hazard.
+ Cartridges for H_2S, CO and flammables (LEL); calibration is a cartridge swap.

CONS

– One gas per unit: you need 3.
– I found no built-in sounder: audible alarm needs an external horn on the relays.
– I found no hazardous-area rating: the unit must sit in a non-classified area.
– H_2S sticks to sample tubing (slower, lower readings on long runs), and diesel vapor can condense.
– Pump, filter and tubing upkeep. Built mainly for semiconductor fabs and labs.

Instrumart ↗ Cartridges (Instrumart) ↗ Frontline Safety ↗

Honeywell Sensepoint XRL

NOT RECOMMENDED

Budget fixed transmitter, set up over Bluetooth

Status LED only | 4–20 mA / Modbus RTU | IP66 / NEMA 4X

`≈ $1,325 – $1,560 per head, ×3, + 16% surcharge`

PROS

+ Low price and rugged industrial build.
+ Versions for H_2S, CO, O_2, H_2 and flammables.

CONS

– No LCD screen: levels are read on a phone app.
– No relays or sounder: it only sends a signal to a controller. **Without a central panel it can't sound an alarm.**
– Catalytic-bead flammable sensor; H_2S poisons it and it can read low.

Instrumart ↗ e-Controls (H_2S) ↗ Frontline Safety ↗

Bottom line: the Sensepoint XRL doesn't qualify (no screen, no local alarm). The Midas S2 only makes sense for sampling a spot where you can't mount a detector. For standalone wall

detectors, stick with the **Sensepoint XCD** (IR version for flammables), the **Crowcon Xgard Bright** or the **SensAlarm FLEX**.

Side by side

OPTION	TYPE	SCREEN	STROBE / HORN	CLASSIFIED AREA	APPROX. COST FOR ALL 3 GASES
2 × Sensidyne SensAlarm FLEX	All-in-one, fixed (2 sensors per unit)	7" color	Built in (95 dB)	Sensors C1D1, display C1D2	Quoted: $13,124 incl. kit and gases
Industrial Scientific Radius BZ1	All-in-one, battery	Yes	Built in (108 dB)	Yes	$7.4k – $14.1k
Honeywell BW RigRat	All-in-one, battery	Yes	Built in	Yes	$9.4k – $11k
Blackline G7 EXO	All-in-one, battery + cell	Yes	Built in	Yes	Subscription
Forensics FD-60-4GAS	All-in-one, fixed	Check	Relay → external	No	≈ $1.4k
3 × Crowcon Xgard Bright	Per gas, fixed	Yes	Built-in option	Yes	≈ $2.9k – $3.9k (quoted: $4,651 incl. calibration kit)
2 × MSA Ultima X5000	Per gas (2 per head)	Yes	Relay → external	Yes	≈ $7.8k – $13k + horns
3 × Honeywell Sensepoint XCD	Per gas, fixed	Backlit LCD	Relay → external	Yes	≈ $6.7k – $9.3k + horns
3 × CTI GG (EXP)	Per gas, fixed	Yes	CTI horn/strobe	EXP version	≈ $4k – $7k incl. horns

OPTION	TYPE	SCREEN	STROBE / HORN	CLASSIFIED AREA	APPROX. COST FOR ALL 3 GASES
3 × Macurco 6-series	Per gas, fixed	Yes	Buzzer + relay → strobe	No	≈ $1.85k + strobes

Things that matter for this install

H_2S kills catalytic LEL sensors

The common "catalytic bead" LEL sensor gets poisoned by H_2S and silicones and slowly stops reading. With H_2S on site, choose an **infrared (IR) LEL** sensor (Crowcon IR, MSA XIR+, Blackline LEL-IR, Radius IR). IR doesn't see hydrogen, but it does see gasoline and most fuel vapors.

Diesel and heavy fuels

Gasoline vapor is easy to detect as % LEL. Diesel, kerosene and jet fuel give off little vapor, so an LEL sensor may read near zero even when the smell is strong. If heavy fuels are the main concern, add a **PID** sensor (an option on the Radius BZ1 and SensAlarm FLEX).

Mounting height

- **H_2S** and **gasoline vapor** are heavier than air: mount 30–45 cm (12–18 in) above the floor, near pits, drains and low spots.
- **CO** is close to air weight: mount in the breathing zone, about 1.2–1.8 m (4–6 ft).
- That's a big reason to prefer one head per gas over a single box.

Typical alarm levels

- H_2S: low 10 ppm / high 15–20 ppm
- CO: low 25 ppm / high 50 ppm
- Fuel vapor: low 10% LEL / high 20% LEL

Set yours with your safety officer and local rules (OSHA, or your country's equivalent).

Running cost

Plan for a bump test with gas every month or so, a full calibration about every 6 months, and new H_2S/CO sensors every 2–3 years. Calibration gas and a regulator cost a few hundred dollars a year. A cheap unit with expensive or hard-to-find sensors costs more in the long run.

Before you order

- Area classification (hazardous or not)
- Power at each point (24 VDC, 120/240 VAC, or battery)
- Gas ranges: H_2S 0–50 ppm, CO 0–500 ppm, LEL 0–100%
- Indoor or outdoor (IP rating, temperature)
- Ask for the horn/strobe in the same quote

Glossary

The technical terms used on this page and in the quotes, in plain words.

H_2S (hydrogen sulfide)

Toxic gas that smells like rotten eggs. At high levels you stop smelling it because it deadens the sense of smell, which is why you need a detector. It is heavier than air and collects in low spots.

CO (carbon monoxide)

Toxic gas with no smell or color, produced by engines, boilers and any incomplete combustion. It weighs about the same as air.

ppm (parts per million)

Unit for measuring toxic gases in small amounts. 10 ppm of H_2S means 10 parts of H_2S per million parts of air.

LEL / % LEL (lower explosive limit)

The lowest concentration of a gas or vapor in air that can ignite. A reading of 20% LEL means there is one fifth of what's needed for an explosion.

Fuel vapors

The gas given off by gasoline, diesel and other liquid fuels. Measured as combustible gas in % LEL, or with a PID sensor for heavy fuels.

Electrochemical sensor

The sensor type normally used for H_2S and CO. Accurate and inexpensive, but it wears out and needs replacing every 2–3 years.

Catalytic (catalytic bead) sensor

Combustible-gas sensor that "burns" a tiny sample to measure it. Cheap and common, but H_2S and silicones poison it and it stops reading correctly without warning.

Infrared (IR) sensor

Combustible-gas sensor that measures how much infrared light the gas absorbs. H_2S doesn't poison it and it lasts longer, but it costs more and can't detect hydrogen.

PID (photoionization detector)

Sensor that measures organic vapors in ppm using an ultraviolet lamp. Works for diesel, kerosene and other heavy fuels an LEL sensor barely sees. The lamp needs periodic cleaning.

Sensor poisoning

Permanent damage to a sensor from exposure to certain chemicals (H_2S, silicones, lead). The sensor stays on but reads lower than reality.

Fixed detector

Personal (portable) monitor

Unit installed permanently on a wall or post, powered by the plant. It watches an area 24 hours a day.

Small detector worn by a worker, with a rechargeable battery. It protects the person wherever they go, but doesn't watch a fixed area.

Transportable area monitor

Battery unit, larger than a personal one, placed at a spot for days or weeks and movable. Examples: Radius BZ1, RigRat.

Sampling (extractive) detector

Unit that pumps air through tubing from the point being watched. The detector itself stays in a safe area. Example: Midas S2.

Head / transmitter

The part of a fixed detector that holds the sensor. A "transmitter" only sends the reading to other equipment; it may have no display or alarm of its own.

Controller / central panel

Panel that receives signals from several detectors and triggers the alarms. This site won't have one, so each detector must alarm on its own.

Standalone

A detector that works alone: it measures, shows the level and triggers its own alarm without a controller.

Relay

A switch inside the detector that trips at an alarm level. Used to turn on a horn, a strobe or a fan.

Horn / strobe

The audible alarm (horn, measured in dB) and the flashing light (strobe or beacon). Some detectors have them built in; others need an external one wired to the relay.

dB (decibels)

How loud a horn is. 95 dB is clearly heard in a workshop; 108 dB carries over a noisy site.

Alarm levels (low / high)

The readings at which the detector warns. Low is a caution; high means evacuate or act immediately.

4–20 mA / Modbus

Standard ways to send a detector's reading to a controller or control system. Not needed for standalone detectors, but useful if a controller is added later.

Classified (hazardous) area

A zone where flammable gas or vapor can be in the air. Only equipment certified not to cause a spark can be installed there.

Class I, Division 1 / Division 2

The US classification for flammable-gas areas. Division 1: gas may be present in normal operation. Division 2: only during a

leak or failure. Set by an engineer or local code.

Zone 0 / 1 / 2 (ATEX / IECEx)

The equivalent international classification. Zone 1 roughly matches Division 1, and Zone 2 matches Division 2.

Explosion-proof (XP / Ex d)

A housing built so that if a spark happens inside, the explosion can't escape. Required in classified areas.

General purpose

Equipment with no hazardous-area rating. Only for non-classified zones such as offices, workshops or warehouses without flammable vapors.

IP66 / NEMA 4X

Protection ratings against dust and water. IP66 and NEMA 4X mean it withstands water jets and outdoor use.

Bump test

A quick test that exposes the detector to a small amount of gas to confirm it responds and the alarm sounds. Done periodically.

Calibration

Adjusting the detector with a gas of known concentration so the reading is accurate. Usually every 6 months.

Calibration gas

A cylinder with an exact gas mixture, used for bump tests and calibration. You need one per sensor type.

Calibration kit

The accessories for calibrating: cylinder regulator, tubing, and the adapter or cup that fits over the sensor.

Sensor cartridge

The replacement sensor swapped in when the original reaches end of life, without replacing the whole detector.

Surcharge

An extra charge on top of list price. Honeywell added 16% to many of its products in 2025.

Incoterms

International terms that define who pays shipping, insurance and customs, and where risk passes to the buyer (for example, EXW or DAP).

Research compiled October 2026 from manufacturer pages and online reseller listings. Prices and availability change; get a written quote from an authorized distributor. This page compares products; it is not an engineering design. A qualified installer should confirm the area classification, placement and alarm set points.